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Transfer apparatus

Abstract

With reference to a plurality of images formed by an imaging unit, the position of a robot hand to be inserted into a cassette at the time of unloading a workpiece from the cassette is determined. Hence, it is not necessary for an operator to recognize in advance configuration details of the cassette and a warp or bending of the workpiece accommodated in the cassette. Accordingly, the operator's work at the time of unloading the workpiece from the cassette is simplified.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a transfer apparatus that transfers a workpiece.

Description of the Related Art

(2) Chips having devices such as integrated circuits (ICs) formed thereon are essential components in various kinds of electronic equipment such as mobile phones and personal computers. Such chips are manufactured by, for example, dividing a workpiece such as a wafer having a front surface on which multiple devices are formed, into regions including respective devices.

(3) Workpieces having devices formed thereon are often thinned before being divided, for the purpose of reduction in size and weight of chips, for example. Examples of a method for thinning a workpiece include grinding carried out by a grinding apparatus including a chuck table for holding a front surface (lower surface) side of the workpiece and a grinding wheel that is disposed above the chuck table and has a plurality of grinding stones arranged in an annular and discrete manner.

(4) In order to grind the workpiece in this case, as a premise, the workpiece needs to be loaded onto

the grinding apparatus. In a general chip manufacturing process, for example, workpieces corresponding to one lot (substantially 25 workpieces) are accommodated in a cassette, and this cassette is loaded onto a placement surface of a cassette table provided in the grinding apparatus. Further, the grinding apparatus generally has a transfer robot (unloading means) that uses its robot hand to hold the workpiece and unload the workpiece from the cassette (refer to Japanese Patent Laid-open No. 2001-284303, for example).

SUMMARY OF THE INVENTION

(5) The cassette generally has a pair of side walls that face each other and are formed with a plurality of grooves on respective inner side surfaces. A workpiece such as a wafer is accommodated in the cassette such that an outer peripheral edge of the workpiece is disposed in one of the plurality of grooves of each side wall. Meanwhile, configuration details of the cassette including a distance between the pair of side walls and an interval for arranging the plurality of grooves often vary depending on the size of workpieces to be accommodated.

(6) Hence, when the transfer robot is used to unload a workpiece from the cassette, an operator is required to recognize the configuration details of the cassette in advance and input, into the transfer robot, information for setting an appropriate position of the robot hand to be inserted into the cassette. However, in a case in which many different kinds of cassettes have different configuration details, it is troublesome for the operator to recognize the configuration details of each cassette.

(7) Moreover, even when cassettes of the same kind are used, dimensional errors may occur in configuration details of the cassettes due to a difference in manufacturer, production lot, or the like. In such a case, when the transfer robot is used to unload a workpiece from a cassette, even if the operator makes reference to information that the operator can get in advance, it may be difficult to set an appropriate position of the robot hand to be inserted into the cassette.

(8) In addition, due to a warp or bending of a workpiece accommodated in a cassette, the appropriate position of the robot hand to be inserted into the cassette to unload the workpiece from the cassette may change. To cope with such a case, the operator is required to find the appropriate position of the robot hand to be inserted into the cassette on each occasion, which leads to complex work.

(9) In view of the problems described above, it is an object of the present invention to provide a transfer apparatus that can simplify the operator's work required when a workpiece is unloaded from the cassette.

(10) In accordance with an aspect of the present invention, there is provided a transfer apparatus including a cassette table, a transfer robot, and a controller. The cassette table has a placement surface on which a cassette that has a pair of side walls facing each other and accommodates a workpiece is placed, the pair of side walls having respective inner side surfaces on each of which a plurality of grooves are formed, the workpiece being accommodated with its outer peripheral edge disposed in one of the grooves of each of the side walls. The transfer robot unloads the workpiece from the cassette. The controller controls operation of the transfer robot. The transfer robot includes a transfer arm having a tip end that is movable in a first direction perpendicular to the placement surface and a second direction parallel to the placement surface, a robot hand that is disposed at the tip end of the transfer arm and is capable of holding the workpiece, and an imaging unit that is fixed to the transfer arm and is capable of imaging a region included in the cassette placed on the placement surface. The controller causes the imaging unit to image different regions of the cassette accommodating the workpiece to form a plurality of images, and thereafter determines a position in the first direction and a position in the second direction of the robot hand to be inserted into the cassette at the time of unloading the workpiece from the cassette, with reference to the plurality of images.

(11) Preferably, the plurality of images include a side-wall image formed by imaging a region containing two or more grooves adjacent in the first direction among the plurality of grooves, and the controller measures an interval for arranging the plurality of grooves in the first direction, with

reference to the side-wall image, and thereafter determines the position in the first direction of the robot hand at the time of unloading the workpiece from the cassette, with reference to the measured interval.

(12) In the present invention, with reference to a plurality of images formed by imaging by an imaging unit, the position of a robot hand to be inserted into a cassette at the time of unloading a workpiece from the cassette is determined. Hence, in the present invention, it is not necessary for an operator to recognize in advance configuration details of the cassette and a warp or bending of the workpiece accommodated in the cassette. Accordingly, the operator's work at the time of unloading the workpiece from the cassette is simplified.

(13) The above and other objects, features and advantages of the present invention and the manner of realizing them will become more apparent, and the invention itself will best be understood from a study of the following description and appended claims with reference to the attached drawings showing a preferred embodiment of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view schematically illustrating an example of a grinding apparatus including a transfer apparatus that transfers a workpiece;
- (2) FIG. 2 is a perspective view schematically illustrating an example of a transfer robot;
- (3) FIG. 3 is a perspective view schematically illustrating an example of the workpiece;
- (4) FIG. 4 is a perspective view schematically illustrating an example of a cassette;
- (5) FIG. 5 is a plan view schematically illustrating configurations of a turn table and peripheral parts thereof;
- (6) FIG. 6A is a perspective view schematically illustrating a state in an example of operation of the transfer robot that is to unload a workpiece from the cassette;
- (7) FIG. 6B is a perspective view schematically illustrating another state in the example of operation of the transfer robot that is to unload a workpiece from the cassette; and
- (8) FIG. 6C is a perspective view schematically illustrating yet another state in the example of operation of the transfer robot that is to unload a workpiece from the cassette.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(9) A preferred embodiment of the present invention will be described hereinbelow with reference to the accompanying drawings. FIG. 1 is a perspective view schematically illustrating an example of a grinding apparatus including a transfer apparatus that transfers a workpiece. It is to be noted that, in FIG. 1, some components of the grinding apparatus are illustrated as functional blocks. Further, an X-axis direction (front-rear direction) and a Y-axis direction (left-right direction) indicated in FIG. 1 are directions perpendicular to each other in a horizontal plane, and a Z-axis direction (up-down direction or height direction) is a direction (vertical direction) perpendicular to both the X-axis direction and the Y-axis direction.

(10) The grinding apparatus, denoted by 2, in FIG. 1 includes a base 4 that supports various components. The base 4 has an opening 4a defined on a front-end side of an upper surface thereof. In the opening 4a, there are provided a transfer robot 6 and a Y-axis movement mechanism (not illustrated) for moving the transfer robot 6 along the Y-axis direction. The Y-axis movement mechanism includes, for example, a ball screw, a motor, and the like.

(11) FIG. 2 is a perspective view schematically illustrating the transfer robot 6. The transfer robot 6 includes a first drive portion 8 in a columnar shape extending in the Z-axis direction. The first drive portion 8 has its lower portion fixed to the Y-axis movement mechanism. Inside the first drive portion 8, there is provided an actuator (not illustrated) such as an air cylinder that has a piston rod movable along the Z-axis direction and that can rotate about its rotational axis extending along the

Z-axis direction.

(12) Further, the first drive portion **8** has an opening defined at an upper surface side thereof for allowing the piston rod to pass through. To an upper end portion of the piston rod, a transfer arm **10** is coupled. Hence, the transfer arm **10** is movable along the Y-axis direction and the Z-axis direction. That is, when the Y-axis movement mechanism is actuated, the transfer arm **10** moves along the Y-axis direction together with the first drive portion **8**. When the actuator provided inside the first drive portion **8** is energized, the transfer arm **10** moves along the Z-axis direction (moves up and down).

(13) The transfer arm **10** is a robot arm having a plurality of joints. More specifically, the transfer arm **10** has a first arm portion **10a** in a plate shape extending in a direction perpendicular to the Z-axis direction. The first arm portion **10a** is coupled at a lower side of one end portion thereof to the upper end portion of the piston rod in such a manner that the first arm portion **10a** moves and rotates together with the piston rod. The first arm portion **10a** is coupled at an upper side of the other end portion thereof to a lower side of a columnar first joint portion (not illustrated).

(14) The first joint portion is coupled at an upper side thereof to a second arm portion **10b** in a plate shape extending in a direction perpendicular to the Z-axis direction. The second arm portion **10b** is coupled at a lower side of one end portion thereof to the upper side of the other end portion of the first arm portion **10a** through the first joint portion in such a manner that the second arm portion **10b** can rotate about its rotational axis extending along the Z-axis direction. The second arm portion **10b** is coupled at an upper side of the other end portion thereof to a lower side of a columnar second joint portion **10c**.

(15) The second joint portion **10c** is coupled at an upper side thereof to a second drive portion **10d** in a rectangular parallelepiped shape extending in a direction perpendicular to the Z-axis direction. The second drive portion **10d** is coupled at a lower side of one end portion thereof to the upper side of the other end portion of the second arm portion **10b** through the second joint portion **10c** in such a manner that the second drive portion **10d** can rotate about its rotational axis extending along the Z-axis direction.

(16) Moreover, the second drive portion **10d** has, fixed to a side surface thereof, an imaging unit **12** extending along the same direction as that of the second drive portion **10d**. The imaging unit **12** includes, for example, a light source such as a light emitting diode (LED), an objective lens, and an imaging element such as a charge coupled device (CCD) image sensor or a complementary metal oxide semiconductor (CMOS) image sensor. The imaging unit **12** images a structure existing in, as viewed from the imaging unit **12**, a direction from the other end portion toward the one end portion of the second drive portion **10d** to form an image.

(17) Inside the second drive portion **10d**, there is provided a motor (not illustrated) for rotating a spindle **10e** about its rotational axis extending along a direction perpendicular to the Z-axis direction. The spindle **10e** extends through an opening defined at a surface of the second drive portion **10d** on the other end portion side and has its tip end portion exposed outward. To the tip end portion of the spindle **10e**, a base end portion in a rectangular parallelepiped shape of a robot hand **14** is coupled through a coupling portion **10f** in a plate shape. That is, the robot hand **14** is disposed at a tip end of the transfer arm **10**.

(18) The robot hand **14** further has an elliptic plate portion integrated with the base end portion thereof. More specifically, the elliptic plate portion has such a shape that the major axis of the ellipse extends in parallel to the spindle **10e**, and has a linear cutout formed from a center to a tip end thereof. In one surface of the elliptic plate portion of the robot hand **14**, a plurality of suction holes (not illustrated) are defined.

(19) The suction holes are connected to a suction source (not illustrated) such as an ejector via, for example, a passage provided inside the robot hand **14** or a valve for controlling a flow of gas. When the suction source is actuated with the valve opened, a suction force acts on a space near the one surface of the elliptic plate portion of the robot hand **14**.

(20) Hence, the one surface functions as a holding surface to hold a workpiece under suction. In the transfer robot **6**, when the spindle **10e** is rotated in a state in which a workpiece is held by the holding surface of the robot hand **14**, the workpiece can be turned upside down. In other words, the workpiece can be held on either an upper side or a lower side of the robot hand **14**.

(21) FIG. **3** is a perspective view schematically illustrating an example of a workpiece transferred by the transfer robot **6**. The workpiece, denoted by **11**, in FIG. **3** is a wafer consisting of a semiconductor material such as silicon (Si), for example. The workpiece **11** is partitioned into a plurality of regions by a plurality of projected dicing lines **13** that intersect each other, and devices **15** such as ICs or large-scale integration (LSI) circuits are formed in the respective regions on a front surface **11a** of the workpiece **11**.

(22) It is to be noted that the material, shape, configuration, size, and the like of the workpiece **11** are not limited. For example, the workpiece **11** may be a substrate consisting of another semiconductor material or a material such as ceramics, resin, or metal. Likewise, the kind, quantity, shape, configuration, size, arrangement, and the like of the devices **15** are not limited either.

(23) Further, a film tape having a diameter substantially equal to that of the workpiece **11** may be attached to the front surface **11a** of the workpiece **11**. The tape consists of resin, for example, and protects the devices **15** by mitigating an impact to be applied to the front surface **11a** side of the workpiece **11** when the workpiece **11** is ground on a back surface **11b** side thereof.

(24) FIG. **4** is a perspective view schematically illustrating an example of a cassette accommodating the workpiece **11**. The cassette, denoted by **16**, in FIG. **4** has a top plate **16a** in a flat-plate shape. The top plate **16a** is a rectangular flat plate having four corners, with a pair of adjacent corners of the four corners chamfered and the other pair of corners not chamfered.

(25) On a lower side of an end portion (rear end portion) positioned between the pair of chamfered corners of the top plate **16a**, an upper end portion of a side wall (not illustrated) extending in a direction (height direction) perpendicular to the top plate **16a** is fixed. On respective lower sides of two end portions (left end portion and right end portion) each positioned between a chamfered corner and an unchamfered corner of the top plate **16a**, upper end portions of side walls **16b** and **16c** extending in the height direction are fixed.

(26) Meanwhile, on a lower side of an end portion (front end portion) positioned between the pair of unchamfered corners of the top plate **16a**, no side wall extending in the height direction is fixed. That is, the lower side of the front end portion of the top plate **16a** is opened, serving as an opening surface. On an inner side surface of each of the side walls **16b** and **16c**, a plurality of grooves **16d** each extending in a direction perpendicular to the height direction are formed at predetermined intervals in the height direction.

(27) More specifically, each of the plurality of grooves **16d** formed on the inner side surface of the side wall **16b** faces corresponding one of the plurality of grooves **16d** formed on the inner side surface of the side wall **16c**. Each groove **16d** is substantially rectangular in cross section. In other words, each groove **16d** has a pair of inner side surfaces lying in a direction substantially perpendicular to the height direction and a bottom surface lying substantially in parallel to the height direction.

(28) When the workpiece **11** is accommodated in the cassette **16**, an outer peripheral edge of the workpiece **11** is placed on the inner side surface, which is the farther from the top plate **16a**, of the pair of inner side surfaces of one of the plurality of grooves **16d** of each of the side walls **16b** and **16c**. That is, the outer peripheral edge of the workpiece **11** is supported on the inner side surfaces, which are the farther from the top plate **16a**, of the respective pairs of inner side surfaces of one facing pair of grooves **16d**.

(29) It is to be noted that the number of grooves **16d** formed on the inner side surface of the side wall **16b** and the number of grooves **16d** formed on the inner side surface of the side wall **16c** are not limited. For example, the grooves **16d** corresponding to the number of workpieces **11** of one lot (substantially 25 workpieces) may be formed in the cassette **16**.

(30) Further, a lower portion of the side wall **16b** and a lower portion of the side wall **16c** are coupled to each other through a connection member **16e** in an elongated plate shape. With a plurality of workpieces **11** accommodated therein, for example, the cassette **16** is loaded onto a cassette table **18a** or **18b** of the grinding apparatus **2** illustrated in FIG. **1**.

(31) The cassette tables **18a** and **18b** each have an upper surface (placement surface) extending in parallel to the X-axis direction and the Y-axis direction, and a prism-shaped cassette stopper **19** extending along the Y-axis direction is provided at an end portion of the placement surface on the opening **4a** side. The cassette **16** is placed on the placement surface of the cassette table **18a** or **18b** in such a manner that the opening surface of the cassette **16** is positioned on the opening **4a** side and lower end portions of the side walls **16b** and **16c** make contact with the cassette stopper **19**.

(32) In this case, outer side surfaces of the side walls **16b** and **16c** of the cassette **16** lie in parallel to the X-axis direction. This prevents the robot hand **14** of the transfer robot **6** from bumping against the inner side surfaces of the side walls **16b** and **16c** when the robot hand **14** is inserted into the cassette **16**.

(33) Further, on an obliquely rear side of the transfer robot **6**, there is provided a position adjustment mechanism **20** for adjusting the position of the workpiece **11**. The position adjustment mechanism **20** includes, for example, a disk-shaped table **20a** having an upper surface (support surface) that can support the workpiece **11** thereon and a plurality of pins **20b** arranged around the table **20a**.

(34) When the workpiece **11** unloaded by the transfer robot **6** from the cassette **16** is loaded onto the support surface of the table **20a**, the plurality of pins **20b** move in a radial direction of the table **20a** in such a manner as to make contact with the outer peripheral edge of the workpiece **11**. The workpiece **11** thus has its center aligned with a predetermined position in the X-axis direction and the Y-axis direction. It is to be noted that details of unloading of the workpiece **11** from the cassette **16** carried out by the transfer robot **6** will be described later.

(35) On an obliquely rear side of the position adjustment mechanism **20** (on a rear side of the transfer robot **6**), there is provided a transfer mechanism **22** that holds the workpiece **11** and transfers the workpiece **11** rearward. The transfer mechanism **22** includes a holding pad for holding the workpiece **11** under suction and an arm connected to the holding pad. The transfer mechanism **22** transfers the workpiece **11** having been adjusted in position by the position adjustment mechanism **20**, rearward by turning the holding pad with use of the arm.

(36) On a rear side of the transfer mechanism **22**, there is provided a turn table **24** in a disk shape. The turn table **24** is connected to a rotational drive source (not illustrated) such as a motor and rotates about its rotational axis extending in substantially parallel to the Z-axis direction. The turn table **24** has, on an upper surface thereof, three chuck tables **26** each for holding the workpiece **11** thereon.

(37) The three chuck tables **26** are arranged substantially at equal intervals along a circumferential direction of the turn table **24**. It is to be noted that the number of chuck tables **26** arranged on the turn table **24** is not limited. FIG. **5** is a plan view schematically illustrating configurations of the turn table **24** and peripheral parts thereof.

(38) It is to be noted that, in FIG. **5**, for convenience of explanation, some components are indicated by broken lines. The transfer mechanism **22** loads the workpiece **11** held by the holding pad onto the chuck table **26** positioned in a loading/unloading area A (refer to FIG. **5**) adjacent to the transfer mechanism **22**. The turn table **24** rotates, for example, in a direction indicated by an arrow in FIG. **1** and FIG. **5**, and each of the chuck tables **26** moves to the loading/unloading area A, a rough-grinding area B, and a finish-grinding area C in this order.

(39) Each of the chuck tables **26** is connected to a rotational drive source (not illustrated) such as a motor and rotates about its rotational axis extending in substantially parallel to the Z-axis direction. The chuck table **26** has a disk-shaped frame body consisting of a metallic material such as stainless steel, for example. On an upper surface side of the frame body, there is formed a recessed portion

having a circular opening at its upper end. A disk-shaped porous plate consisting of ceramics or the like is fixed in the recessed portion.

(40) The chuck table **26** has an upper surface formed into a shape corresponding to a side surface of a cone that has its center slightly protruding compared to its outer edge, and the upper surface functions as a holding surface **26a** for holding the workpiece **11** thereon. That is, the chuck table **26** has the holding surface **26a** for holding the workpiece **11** thereon, at an upper portion thereof.

(41) The holding surface **26a** is connected to a suction source (not illustrated) such as an ejector through a suction passage (not illustrated) or the like formed inside the chuck table **26**. When the suction source is actuated, a suction force acts on a space near the holding surface **26a**, so that the workpiece **11** having been loaded onto the chuck table **26** is held under suction.

(42) As illustrated in FIG. **1**, two columnar support structures **28** are provided on respective rear sides of the rough-grinding area B and the finish-grinding area C (on a rear side of the turn table **24**). The support structures **28** each have a Z-axis movement mechanism **30** provided on a front surface side thereof. The Z-axis movement mechanism **30** has a pair of guide rails **32** that are fixed to the front surface of the support structure **28** and extend along the Z-axis direction.

(43) On a front surface side of the pair of guide rails **32**, a movable plate **34** is coupled in such a manner as to slidably move along the pair of guide rails **32**. Between the pair of guide rails **32**, a screw shaft **36** extending along the Z-axis direction is disposed. To one end portion of the screw shaft **36**, a motor **38** for rotating the screw shaft **36** about its rotational axis is coupled.

(44) A nut (not illustrated) that contains a number of balls rollingly movable in a helical groove formed on a surface of the screw shaft **36** is operatively threaded over the screw shaft **36**, which constitutes a ball screw. When the motor **38** is energized to rotate the screw shaft **36** about its rotational axis, the balls roll in the helical groove and circulate through the nut, causing the nut to move along the Z-axis direction. The nut is fixed to a rear surface (back surface) side of the movable plate **34**.

(45) Accordingly, when the screw shaft **36** is rotated about its rotational axis by the motor **38** coupled to the one end portion of the screw shaft **36**, the movable plate **34** move along the Z-axis direction together with the nut. Further, a fixture **40** is provided to a front surface side of the movable plate **34**. The fixture **40** supports a grinding unit **42** for grinding the workpiece **11**. The grinding unit **42** includes a spindle housing **44** fixed to the fixture **40**.

(46) The spindle housing **44** houses a spindle **46** in such a manner that the spindle **46** can rotate about its rotational axis extending along the Z-axis direction. The spindle **46** has a lower end portion exposed from a lower end surface of the spindle housing **44**. A disk-shaped mount **48** is fixed to the exposed lower end portion of the spindle **46**.

(47) To a lower surface of the mount **48** of the grinding unit **42** on the rough-grinding area B side, a first grinding wheel **50a** for rough grinding is mounted. The first grinding wheel **50a** for rough grinding includes a first wheel base that consists of metal such as stainless steel or aluminum and that has a diameter substantially equal to that of the mount **48**.

(48) On a lower surface of the first wheel base, there are provided a plurality of first grinding stones that are arranged in an annular manner and that are formed by bonding abrasive grains of diamond or the like suitable for rough grinding by using a vitrified bond, a resinoid bond, or other kind of bond. In the spindle housing **44** of the grinding unit **42** on the rough-grinding area B side, a first rotational drive source (not illustrated) such as a motor connected to an upper end side of the spindle **46** is housed.

(49) When the first rotational drive source is actuated to rotate the spindle **46**, the first grinding wheel **50a** rotates about its rotational axis. In the vicinity of the first grinding wheel **50a**, there is provided a liquid supply nozzle (not illustrated) that can supply liquid (grinding liquid) such as pure water to a part (processing point) where the workpiece **11** and the first grinding stones make contact with each other. It is to be noted that, instead of the liquid supply nozzle or in addition to the liquid supply nozzle, a liquid supply port to be used for supplying liquid may be provided to the

first grinding wheel **50a**.

(50) Similarly, to a lower surface of the mount **48** of the grinding unit **42** on the finish-grinding area C side, a second grinding wheel **50b** for finish grinding is mounted. The second grinding wheel **50b** for finish grinding includes a second wheel base that consists of metal such as stainless steel or aluminum and that has a diameter substantially equal to that of the mount **48**.

(51) On a lower surface of the second wheel base, there are provided a plurality of second grinding stones that are arranged in an annular manner and that are formed by bonding abrasive grains of diamond or the like suitable for finish grinding by using a vitrified bond, a resinoid bond, or other kind of bond. In the spindle housing **44** of the grinding unit **42** on the finish-grinding area C side, a second rotational drive source (not illustrated) such as a motor connected to an upper end side of the spindle **46** is housed.

(52) When the second rotational drive source is actuated to rotate the spindle **46**, the second grinding wheel **50b** rotates about its rotational axis. In the vicinity of the second grinding wheel **50b**, there is provided a liquid supply nozzle (not illustrated) that can supply liquid (grinding liquid) such as pure water to a part (processing point) where the workpiece **11** and the second grinding stones make contact with each other. It is to be noted that, instead of the liquid supply nozzle or in addition to the liquid supply nozzle, a liquid supply port to be used for supplying liquid may be provided to the second grinding wheel **50b**.

(53) The workpiece **11** held on each chuck table **26** is successively ground by the two grinding units **42** described above. More specifically, the workpiece **11** held on the chuck table **26** positioned in the rough-grinding area B is ground by the grinding unit **42** on the rough-grinding area B side, and the workpiece **11** held on the chuck table **26** positioned in the finish-grinding area C is ground by the grinding unit **42** on the finish-grinding area C side.

(54) At a position in front of the loading/unloading area A and on a side of the transfer mechanism **22**, there is provided another transfer mechanism **52** that holds the workpiece **11** having been ground and transfers the workpiece **11** forward. The transfer mechanism **52** includes a holding pad for holding the workpiece **11** under suction and an arm connected to the holding pad. The transfer mechanism **52** transfers the workpiece **11** having been ground, forward from the chuck table **26** by turning the holding pad with use of the arm.

(55) In front of the transfer mechanism **52**, there is provided a cleaning unit **54** that cleans the workpiece **11** transferred thereto by the transfer mechanism **52**. The cleaning unit **54**, for example, includes a spinner table **54a** that has an upper surface (support surface) capable of supporting the workpiece **11** thereon and that spins with the workpiece **11** held on the support surface, and a nozzle (not illustrated) that ejects a cleaning fluid toward the workpiece **11** held by the spinner table **54a**.

(56) The workpiece **11** having been cleaned by the cleaning unit **54** is unloaded from the spinner table **54a** by the transfer robot **6** and loaded into the cassette **16**. Operation of the respective components of the grinding apparatus **2** is controlled by a controller **56** incorporated in the grinding apparatus **2**. The controller **56** includes, for example, a processing section **58** that controls the respective components of the grinding apparatus **2** and a storage section **60** that stores therein various kinds of information (data, programs, and the like) used in the processing section **58**.

(57) The storage section **60** stores therein, for example, a width (length in the Y-axis direction) of the placement surfaces of the cassette tables **18a** and **18b**. The function of the processing section **58** is implemented, for example, by a central processing unit (CPU) that reads and executes programs stored in the storage section **60**. The function of the storage section **60** is implemented by at least one of a semiconductor memory such as a dynamic random access memory (DRAM), a static random access memory (SRAM), or a NAND flash memory and a magnetic storage device such as a hard disk drive (HDD).

(58) The grinding apparatus **2** may further include components other than those described above. For example, the grinding apparatus **2** may include a touch panel that has a touch sensor for

inputting an instruction by the operator to the controller **56** and a display for outputting various kinds of information for the operator.

(59) It is to be noted that the grinding apparatus **2** illustrated in FIG. **1** can also be expressed as including a transfer apparatus that mainly has the transfer robot **6**. More specifically, the transfer apparatus includes the cassette tables **18a** and **18b** each having the placement surface on which the cassette **16** is to be placed, the transfer robot **6** that unloads the workpiece **11** from the cassette **16**, and the controller **56** that controls operation of the transfer robot **6**.

(60) An example of operation of the transfer robot **6** at the time of unloading the workpiece **11** from the cassette **16** will be described hereinbelow with reference to FIG. **6A**, FIG. **6B**, and FIG. **6C**. In this operation, first, the cassette **16** is placed on the placement surface of the cassette table **18a** or **18b** in such a manner that the opening surface of the cassette **16** is positioned on the opening **4a** side and the lower end portions of the side walls **16b** and **16c** make contact with the cassette stopper **19**. Then, the imaging unit **12** is positioned at such a position that the imaging unit **12** can image one end of the placement surface of the cassette table **18a** or **18b** and a space right above the one end.

(61) More specifically, the processing section **58** causes the Y-axis movement mechanism provided inside the opening **4a** to be actuated in such a manner that the transfer robot **6** is positioned on a rear side of one end of the placement surface of the cassette table **18a** or **18b**. Then, the processing section **58** controls the actuator provided inside the first drive portion **8** of the transfer robot **6** to move the imaging unit **12** and the like up and down and cause the first arm portion **10a**, the second arm portion **10b**, and/or the second drive portion **10d** to rotate about their respective rotational axes, in such a manner that the objective lens of the imaging unit **12** faces forward and is positioned at a height substantially equal to that of the placement surface of the cassette table **18a** or **18b**.

(62) Next, the processing section **58** causes the imaging unit **12** to image the one end of the placement surface of the cassette table **18a** or **18b** and the space right above the one end (refer to FIG. **6A**). When an image (first image) formed by this imaging contains the side wall **16c** of the cassette **16**, the processing section **58** measures an interval (first interval: **I1**) in the Y-axis direction between the one end of the placement surface of the cassette table **18a** or **18b** and the side wall **16c** of the cassette **16** on the basis of the first image.

(63) When the first image does not contain the side wall **16c** of the cassette **16**, on the other hand, the processing section **58** causes the transfer robot **6** to move along the Y-axis direction by a predetermined distance in such a manner as to move toward another end of the placement surface of the cassette table **18a** or **18b**. It is to be noted that the predetermined distance is set, for example, to a value substantially equal to a width (length in the Y-axis direction) of a space contained in an image formed by the imaging by the imaging unit **12**.

(64) Then, the processing section **58** causes the imaging unit **12** to image a portion of the placement surface of the cassette table **18a** or **18b** and a space right above the portion. When an image (second image) formed by this imaging contains the side wall **16c** of the cassette **16**, the processing section **58** measures the first interval **I1** on the basis of the first image and the second image.

(65) When the second image does not contain the side wall **16c** of the cassette **16**, on the other hand, movement of the transfer robot **6** and the imaging by the imaging unit **12** described above are repeatedly carried out until an image containing the side wall **16c** of the cassette **16** is formed, and thereafter, the processing section **58** measures the first interval **I1**.

(66) Subsequently, the imaging unit **12** is positioned at such a position that the imaging unit **12** can image the other end of the placement surface of the cassette table **18a** or **18b** and a space right above the other end. More specifically, the processing section **58** causes the Y-axis movement mechanism provided inside the opening **4a** to be actuated in such a manner that the transfer robot **6** is positioned on a rear side of the other end of the placement surface of the cassette table **18a** or **18b**.

(67) Next, the processing section **58** causes the imaging unit **12** to image the other end of the placement surface of the cassette table **18a** or **18b** and the space right above the other end (refer to FIG. **6B**). When an image (third image) formed by this imaging contains the side wall **16b** of the cassette **16**, the processing section **58** measures an interval (second interval: **I2**) in the Y-axis direction between the other end of the placement surface of the cassette table **18a** or **18b** and the side wall **16b** of the cassette **16** on the basis of the third image.

(68) When the third image does not contain the side wall **16b** of the cassette **16**, on the other hand, movement of the transfer robot **6** toward the one end of the placement surface of the cassette table **18a** or **18b** and the imaging by the imaging unit **12** are repeatedly carried out until an image (fourth image) containing the side wall **16b** of the cassette **16** is formed. After the fourth image is formed, the processing section **58** measures the second interval **I2** on the basis of the third image and the fourth image.

(69) Subsequently, the processing section **58** specifies the position of a center of the cassette **16** in the Y-axis direction on the basis of the first interval **I1** and the second interval **I2**. Where the width, which is stored in the storage section **60**, of the placement surface of the cassette table **18a** or **18b** is denoted by **W**, for example, the position of the center is a position separated from the one end of the placement surface of the cassette table **18a** or **18b** by a distance of $\{W+(I1-I2)\}/2$.

(70) The processing section **58** causes data representing the position of the center to be stored in the storage section **60** as data representing the position, in the Y-axis direction, of the robot hand **14** to be inserted into the cassette **16** at the time of unloading the workpiece **11** from the cassette **16**.

(71) Then, the imaging unit **12** is positioned at such a position that the imaging unit **12** can image a center of the placement surface of the cassette table **18a** or **18b** and a space right above the center. More specifically, the processing section **58** causes the Y-axis movement mechanism provided inside the opening **4a** to be actuated in such a manner that the transfer robot **6** is positioned on a rear side of the center of the placement surface of the cassette table **18a** or **18b**.

(72) The processing section **58** then causes the imaging unit **12** to image the center of the placement surface of the cassette table **18a** or **18b** and the space right above the center (refer to FIG. **6C**). When an image (fifth image) formed by this imaging contains one or a plurality of workpieces **11**, the processing section **58** measures a height of each of the one or plurality of workpieces **11** from the placement surface of the cassette table **18a** or **18b**, on the basis of the fifth image.

(73) When the fifth image does not contain the top plate **16a** of the cassette **16**, the processing section **58** causes the actuator provided inside the first drive portion **8** of the transfer robot **6** to raise the imaging unit **12** and the like by a predetermined height. It is to be noted that the predetermined height is set, for example, to a value equal to a height (length in the Z-axis direction) of a space contained in an image formed by the imaging by the imaging unit **12**.

(74) Subsequently, the processing section **58** causes the imaging unit **12** to image the space right above the center of the placement surface of the cassette table **18a** or **18b**. When an image (sixth image) formed by this imaging contains one or a plurality of workpieces **11**, the processing section **58** measures a height of each of the one or plurality of workpieces **11** from the placement surface of the cassette table **18a** or **18b**, on the basis of the fifth image and the sixth image.

(75) When the sixth image does not contain the top plate **16a** of the cassette **16**, raising of the imaging unit **12** and the like and the imaging by the imaging unit **12** are repeatedly carried out until an image containing the top plate **16a** of the cassette **16** is formed. When images formed by this repetition of imaging contain one or a plurality of workpieces **11**, the processing section **58** measures a height of each of the one or plurality of workpieces **11** from the placement surface of the cassette table **18a** or **18b**.

(76) Then, the processing section **58** causes data representing a position obtained by offsetting the measured height of each of the one or plurality of workpieces **11** to be stored in the storage section **60** as data representing the position, in the Z-axis direction, of the robot hand **14** to be inserted into

the cassette **16** at the time of unloading each workpiece **11** from the cassette **16**. It is to be noted that this offsetting aims at preventing the robot hand **14** from bumping against the workpiece **11** when the robot hand **14** is inserted into the cassette **16**.

(77) The amount of offsetting is determined, for example, on the basis of an image (side-wall image) containing two or more grooves **16d** adjacent in the Z-axis direction among the plurality of grooves **16d** formed on the inner side surfaces of the side walls **16b** and **16c** of the cassette **16**. More specifically, after measuring the interval for arranging the plurality of grooves **16d** in the Z-axis direction, on the basis of the side-wall image, the processing section **58** determines a distance of one- to three-quarter times the interval (one half of the interval, for example) as the offset amount.

(78) It is to be noted that the side-wall image is formed, for example, by additional imaging carried out in order to determine the offset amount. Alternatively, if any of the first to fourth images described above contains the two or more grooves **16d**, at least one of such images may be utilized as the side-wall image.

(79) Now, one workpiece **11** is unloaded from the cassette **16**. When the workpiece **11** is unloaded from the cassette **16**, first, the robot hand **14** that has been positioned, on the basis of the data stored in the storage section **60**, at a position suitable for unloading the workpiece **11** is inserted into the cassette **16**.

(80) For example, the robot hand **14** is inserted into the cassette **16** in the following manner. First, the processing section **58** causes the spindle **10e** provided inside the second drive portion **10d** to rotate about its rotational axis in such a manner that the holding surface of the robot hand **14** faces upward.

(81) Then, the processing section **58** causes the Y-axis movement mechanism provided inside the opening **4a** to be actuated, with reference to the data that represents the position of the robot hand **14** in the Y-axis direction and that is stored in the storage section **60**. As a result, the transfer robot **6** is positioned on a rear side of the center of the cassette **16** in the Y-axis direction.

(82) Subsequently, the processing section **58** controls the actuator provided inside the first drive portion **8** of the transfer robot **6** to move the imaging unit **12** and the like up and down, with reference to the data that represents the position of the robot hand **14** in the Z-axis direction and that is stored in the storage section **60**. As a result, the holding surface of the robot hand **14** is positioned at a height offset downward by the offset amount described above from the height of the workpiece **11** to be unloaded.

(83) Then, the processing section **58** causes the first arm portion **10a**, the second arm portion **10b**, and/or the second drive portion **10d** to rotate about their respective rotational axes, with reference to the data that represents the position of the robot hand **14** in the Y-axis direction and that is stored in the storage section **60**. As a result, the robot hand **14** is inserted into the cassette **16** in a state in which a center of the holding surface of the robot hand **14** in the Y-axis direction agrees with the center of the cassette **16** in the Y-axis direction.

(84) The processing section **58** then controls the actuator provided inside the first drive portion **8** to raise the robot hand **14** and the like in such a manner that the holding surface of the robot hand **14** makes contact with and supports the workpiece **11**. The processing section **58** then controls the robot hand **14** to cause the workpiece **11** to be held on the robot hand **14** under suction.

(85) Subsequently, the processing section **58** causes the first arm portion **10a**, the second arm portion **10b**, and the second drive portion **10d** to rotate about their respective rotational axes in such a manner that the workpiece **11** is unloaded from the cassette **16**. Accordingly, the unloading of one workpiece **11** from the cassette **16** is completed.

(86) In the above-described transfer apparatus included in the grinding apparatus **2**, the position of the robot hand **14** to be inserted into the cassette **16** at the time of unloading the workpiece **11** from the cassette **16** is determined with reference to the plurality of images formed by imaging by the imaging unit **12**.

- (87) Hence, in the transfer apparatus, it is not necessary for the operator to recognize in advance configuration details of the cassette **16** and a warp or bending of the workpiece **11** accommodated in the cassette **16**. Accordingly, the operator's work at the time of unloading the workpiece **11** from the cassette **16** is simplified.
- (88) Further, in the transfer apparatus included in the grinding apparatus **2**, the workpiece **11** is unloaded from the cassette **16** after the imaging unit **12** images the workpiece **11** accommodated in the cassette **16** to form an image. Hence, in the transfer apparatus, it is possible to readily find an abnormality in an accommodation status of the workpiece **11** in the cassette **16** and an abnormality of the workpiece **11** itself.
- (89) It is to be noted that examples of the abnormality in the accommodation status of the workpiece **11** in the cassette **16** include an obliquely inserted state (state in which the outer peripheral edge of one workpiece **11** is supported in grooves **16d** different in height from each other in the side walls **16b** and **16c**) and a doubly inserted state (state in which the outer peripheral edges of two workpieces **11** are both supported in grooves **16d** of the same height in the side walls **16b** and **16c**).
- (90) Further, examples of the abnormality of the workpiece **11** itself include a state in which the workpiece **11** has a shape unsuitable for grinding (state in which the workpiece **11** is too thick or too thin, or state in which the workpiece **11** has a too large warp or is bent excessively).
- (91) With the grinding apparatus **2**, when an abnormality in the accommodation status of the workpiece **11** or an abnormality of the workpiece **11** itself is found, for example, the operator can be notified of an error message via the touch panel or the like.
- (92) It is to be noted that the details described above are based on one aspect of the present invention, and the present invention is not limited to them. For example, the transfer apparatus included in the grinding apparatus **2** may be provided with one or a plurality of light sources to be used at the time of imaging by the imaging unit **12**.
- (93) In this case, for example, the transfer apparatus may have a light source that emits light from an obliquely upper side, a light source that emits light from an obliquely lower side, and a light source that emits light from an obliquely lateral side toward the cassette **16** placed on the cassette table **18a** or **18b**. In this transfer apparatus, one region of the cassette **16** may be imaged by the imaging unit **12** with the cassette **16** irradiated with light from the different light sources, thereby forming a plurality of images.
- (94) Further, the transfer apparatus may be provided with light sources that can emit light of shape patterns different from each other toward the cassette **16** placed on the cassette table **18a** or **18b**. In this transfer apparatus, one region of the cassette **16** may be imaged by the imaging unit **12** with the cassette **16** irradiated with light of shape patterns different from each other, thereby forming a plurality of images.
- (95) In a case in which one region of the cassette **16** is imaged under different light irradiation conditions in such a manner and a plurality of images are formed, it is easy to recognize a three-dimensional configuration of the workpiece or workpieces **11** included in the region in more details.
- (96) Furthermore, the configuration, method, and the like according to the above-described preferred embodiment may be changed or modified appropriately without departing from the scope of the object of the present invention.
- (97) The present invention is not limited to the details of the above-described preferred embodiment. The scope of the invention is defined by the appended claims and all changes and modifications as fall within the equivalence of the scope of the claims are therefore to be embraced by the invention.

Claims

1. A transfer apparatus comprising: a cassette table having a placement surface on which a cassette that has a pair of side walls facing each other and accommodates a workpiece is placed, the pair of side walls having respective inner side surfaces on each of which a plurality of grooves are formed, the workpiece being accommodated with its outer peripheral edge disposed in one of the grooves of each of the side walls; a transfer robot that unloads the workpiece from the cassette; and a controller that controls operation of the transfer robot, wherein the transfer robot includes: a transfer arm having a spindle which forms a tip end of the transfer arm that is movable in a first direction perpendicular to the placement surface and a second direction parallel to the placement surface, a robot hand that is disposed at the tip end of the transfer arm, such that the robot hand is coupled to the spindle, and is capable of holding the workpiece, and an imaging unit that is fixed to a portion of the transfer arm and is capable of imaging a region included in the cassette placed on the placement surface, and the controller causes the imaging unit to image different regions of the cassette accommodating the workpiece to form a plurality of images, and thereafter determines a position in the first direction and a position in the second direction of the robot hand to be inserted into the cassette at a time of unloading the workpiece from the cassette, with reference to the plurality of images, wherein the controller repeatedly determines whether a reference portion of the cassette is within one of the plurality of images, and when the reference portion of the cassette is not within the one of the plurality of images, the transfer robot is moved a predetermined distance until the imaging unit is positioned such that the image from the imaging unit includes the reference portion of the cassette.
2. The transfer apparatus according to claim 1, wherein the plurality of images include a side-wall image formed by imaging a region containing two or more grooves adjacent in the first direction among the plurality of grooves, and the controller measures an interval for arranging the plurality of grooves in the first direction, with reference to the side-wall image, and thereafter determines the position in the first direction of the robot hand at the time of unloading the workpiece from the cassette, with reference to the measured interval.
3. The transfer apparatus according to claim 1, wherein the transfer robot further comprises a drive portion which houses a portion of the spindle, such that the imaging unit is disposed in contacting relationship with a side wall of the drive portion.
4. The transfer apparatus according to claim 3, wherein the spindle is configured and arranged to rotate about an axis of the spindle.
5. The transfer apparatus according to claim 1, wherein the second direction is also perpendicular to the side walls of the cassette, and wherein the spindle rotates about a spindle axis which is perpendicular to the first direction and the second direction, such that that the robot hand rotates about the spindle axis, wherein the robot hand includes an elliptic plate portion which has a shape where a major axis of the elliptic plate extends parallel to the spindle axis.
6. The transfer apparatus according to claim 5, wherein the elliptic plate portion has a linear cutout formed from a center to a tip end of the elliptic plate portion.
7. The transfer apparatus according to claim 1, wherein the robot hand includes a plurality of suction holes on a holding surface of the robot hand, such that the suction holes on the holding surface hold the workpiece under suction.
8. The transfer apparatus according to claim 1, wherein the spindle is configured and arranged to rotate about an axis of the spindle.
9. A transfer apparatus comprising: a cassette table having a placement surface on which a cassette that has a pair of side walls facing each other and accommodates a workpiece is placed, the pair of side walls having respective inner side surfaces on each of which a plurality of grooves are formed, the workpiece being accommodated with its outer peripheral edge disposed in one of the grooves of each of the side walls; a transfer robot that unloads the workpiece from the cassette; and a controller that controls operation of the transfer robot, wherein the transfer robot includes: a

transfer arm having a tip end that is movable in a first direction perpendicular to the placement surface and a second direction parallel to the placement surface, a robot hand that is disposed at the tip end of the transfer arm and is capable of holding the workpiece, and an imaging unit that is fixed to the transfer arm and is capable of imaging a region included in the cassette placed on the placement surface, and the controller causes the imaging unit to image different regions of the cassette accommodating the workpiece to form a plurality of images, and thereafter determines a position in the first direction and a position in the second direction of the robot hand to be inserted into the cassette at a time of unloading the workpiece from the cassette, with reference to the plurality of images, wherein the controller repeatedly determines whether a reference portion of the cassette is within one of the plurality of images, and when the reference portion of the cassette is not within the one of the plurality of images, the transfer robot is moved a predetermined distance until the imaging unit is positioned such that the image from the imaging unit includes the reference portion of the cassette.
